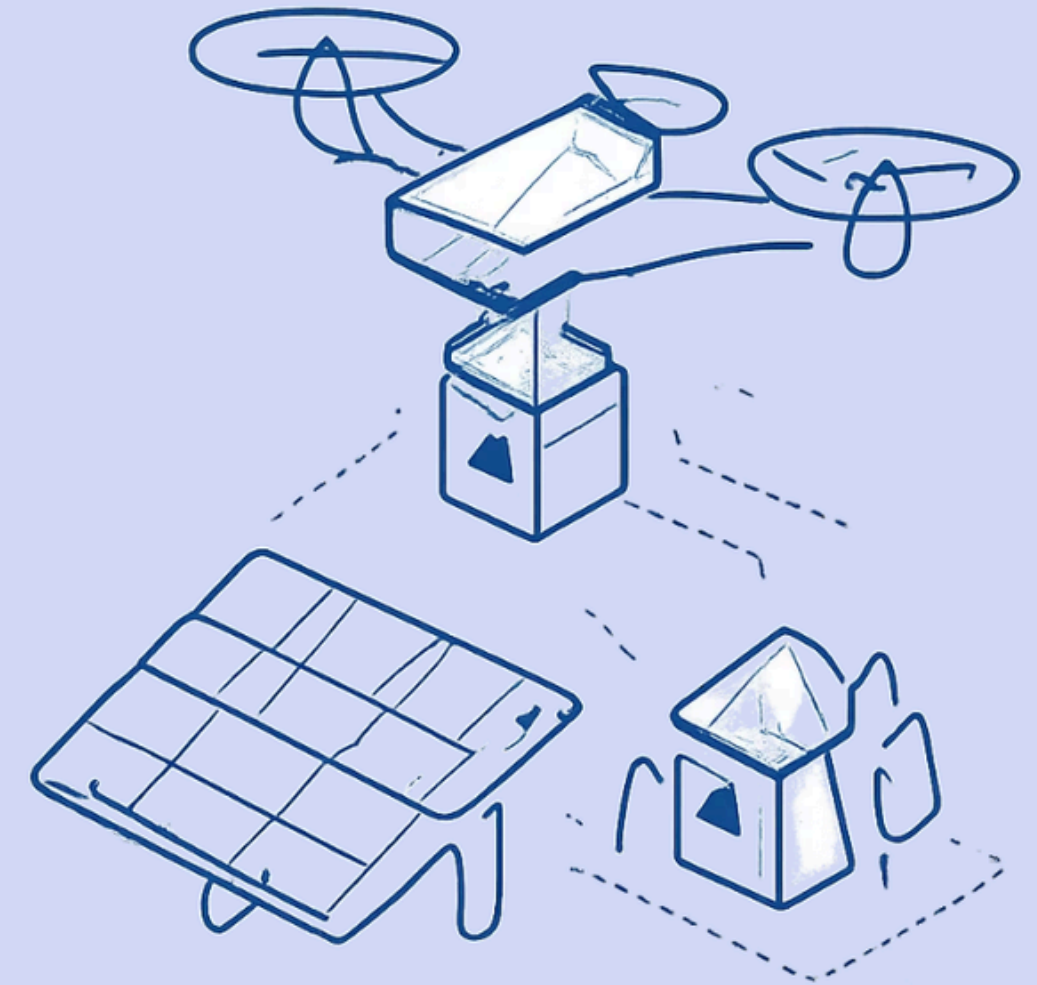




Skyfox – Autonomous Multi-Role UAV Platform with AI Dispatch.

Selected Domain: Robotics, Drones & Smart Automation

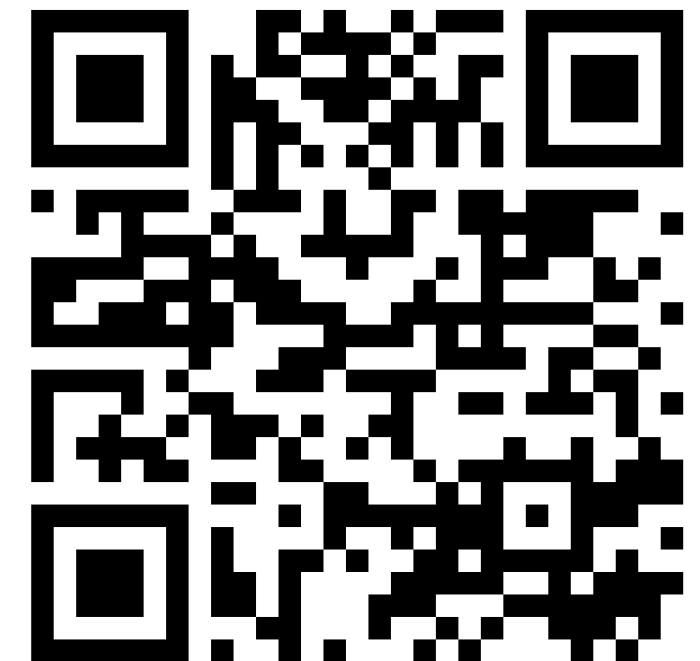
Team Name: RedFox





Simple Question is,

When every second matters, should emergency response wait for humans—or should technology be ready to respond?



<https://arvindtechguy.github.io/skyfox/>



Problem Statement

What problem are you addressing?

- **Disaster Logistics & Safety Hazards:** Manual drone setup wastes critical response time during emergency callouts. Forced ground landings in unstructured debris cause high propeller strike rates and crashes, risking equipment before aid is delivered.
- **Inflexible Single-Purpose Fleet Costs:** Traditional UAVs feature fixed, non-swappable payloads. Purchasing separate drone fleets for search-and-rescue, medical drops, and utility maintenance increases hardware expenditure by over 60%.
- **Solar Maintenance Efficiency & Resource Waste:** Dust buildup reduces solar panel power output by 15 - 30% monthly. Traditional scheduled water cleaning wastes thousands of liters on light surface dust that could be cleared with compressed air.

Who faces this problem?

Emergency response agencies, solar energy operators, and utility companies.

Why is it important?

Time-critical emergency response saves lives; fleet cost reduction enables broader deployment; water efficiency supports renewable energy sustainability.



A Single Drone For Single Purpose

All in one Drone
for Emergency AED,
First Aid, Disaster
Management (Sensing
& Food), Solar
Cleaning, And Custom Slots



Proposed Solution

What is your solution?

Skyfox is an autonomous multi-role drone platform with ML-driven dispatch, modular payload swapping, hover-winch delivery, and conductive fast-charging for continuous **24/7** operation.

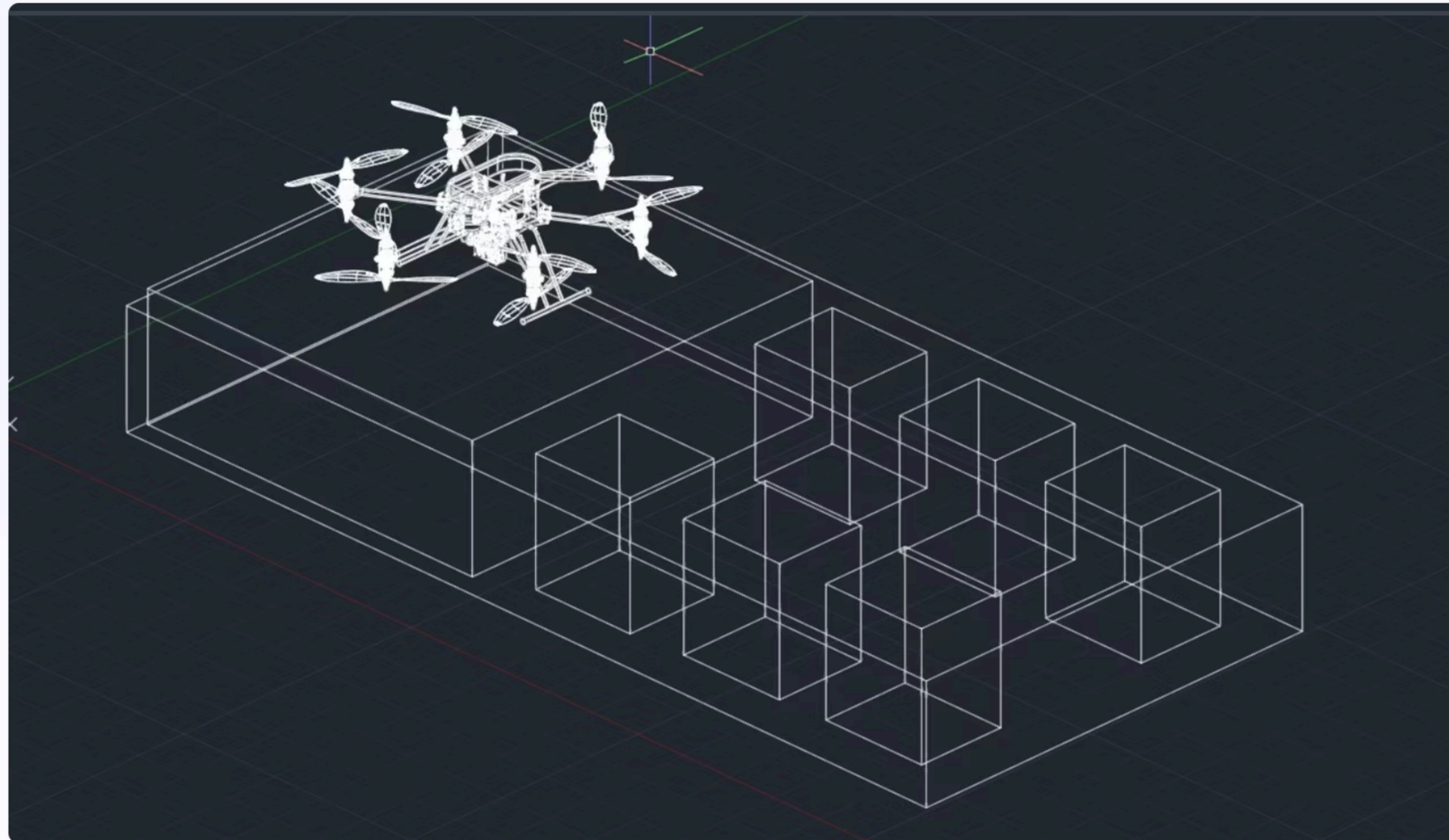
How does it solve the problem?

- **ML-Driven Dispatch Engine** integrates emergency helpline inputs (108/112) to trigger automated payload selection and mechanical auto-latching, eliminating setup delays.
- **Universal Hover-Winch Assembly** lowers payloads safely from stationary hover, eliminating propeller crash hazards on uneven terrain.
- **Conductive Fast-Charging Base Station** enables automated payload swaps for continuous unassisted operation.

Key features / Working concept:

- AED & First-Aid Module for rapid medical supply delivery
- Emergency Rations Module for precision drops to stranded survivors
- Victim Detection Module (ESP32 BLE RSSI + AI vision+ Thermal+ Sonar) for debris rescue
- Solar Cleaning Modules with optical dust-density sensors (80% water savings)

Base Structure (Rack System)





Modular Mission Payload Suite

Four interchangeable payload modules snap onto the universal chassis rack – one drone, every mission.



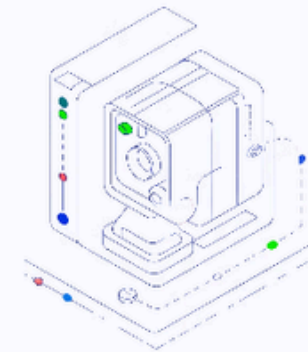
AED & First-Aid Module

Rapid medical supply delivery to victims in inaccessible zones.



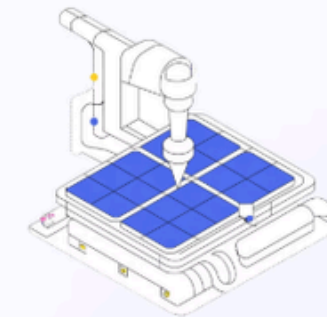
Emergency Rations Module

Precision drop of food and water supplies to stranded survivors.



Victim Detection Module

ESP32 BLE + Thermal+ Sonar + onboard AI vision locates buried or hidden victims in debris.



Solar Cleaning Modules

Optical dust-density sensors select compressed air or targeted water spray – up to 80% water savings.



Innovation & Uniqueness

What makes your idea different?

Skyfox is the first multi-role platform designed specifically for time-critical emergency logistics, bridging emergency response with renewable energy maintenance on a single quadrotor chassis.

Existing solutions:

Traditional UAVs feature fixed, non-swappable payloads requiring separate fleets for different missions. Existing solar maintenance relies on manual labor or water-intensive cleaning. Emergency response drones lack autonomous dispatch integration with emergency services (108/112).

Your competitive advantage:

- First multi-role platform bridging emergency logistics and solar utility maintenance
- Eliminates landing risks via hover-tether winch delivery
- Water-efficient solar maintenance with real-time dust-density detection (80% water savings)
- Zero-human-touch operation with continuous fast-charging and automatic payload latching
- One drone. Every mission. Zero downtime.

Traditional Drones vs Skyfox

Dimension	Traditional Drones	Skyfox	Advantage
Payload Flexibility	X Fixed payloads	✓ Swappable modular payloads	✓
Landing Safety	X Requires landing	✓ Hover-tether winch delivery	✓
Water Efficiency (because Of Dust Density Sensing)	X Water-intensive cleaning	✓ 80% water savings	✓
Operational Autonomy	X Manual dispatch and handling	✓ Zero-human-touch automation	✓



Target Users & Market



Who will use it?

- **Emergency Response Agencies:** Ambulance delays, remote incidents, and time-critical medical delivery make rapid multi-payload drone deployment a life-saving advantage for hospitals, fire departments, and search-and-rescue teams.
- **Solar Energy Operators:** Utility companies and large-scale solar farms managing thousands of panels across distributed sites.
- **Hybrid Market Opportunity:** Organizations operating both emergency services and renewable energy infrastructure can deploy a single unified Skyfox fleet.

Target market:

India's disaster response sector and renewable energy infrastructure operators. Multi-billion-rupee addressable market with strong government tailwinds.

Potential demand:

500+ GW solar capacity expansion + 500+ emergency response agencies across India within 3 years. Potential to serve 10,000+ solar installations.



Feasibility



Technical feasibility

Pixhawk flight controller with optical precision landing, ROS / ESP32 base-station interface, NEMA stepper winch with high-tensile Kevlar tether, and high-current conductive charging plates. All components are commercially available and proven in similar applications.

Resources / Components required

- Pixhawk/ Pixhawk Jetson flight controller
- ROS / ESP32 base-station interface
- NEMA stepper winch with Kevlar tether
- High-current conductive charging plates with hydrophobic sealed enclosure
- Modular payload boxes (AED, rations, detection, cleaning modules)



Approximate cost

₹2.5 - ₹3 Lakhs for complete prototype including master rack, fast-charging pad, and initial payload box suite.

Challenges

- Center-of-gravity balancing during fluid dropouts (solved via dynamic load compensation algorithm)
- Weatherproofing exposed conductive landing contacts (sealed contact enclosure with hydrophobic coating)
- Timeline: 10 - 15 weeks to functional prototype with all four core subsystems operational



Impact & Future Scope

How Many People Could Benefit?

Potential to serve **10,000+ solar installations** and **500+ emergency response agencies** across India within 3 years. Direct impact on disaster response effectiveness and renewable energy efficiency at national scale.

Social / Environmental / Economic Impact

- **Social Impact:** Faster emergency response saves lives in disaster scenarios; victim detection module increases rescue success rates in debris-buried scenarios.
- **Environmental Impact:** 80% reduction in water consumption for solar maintenance; enables renewable energy efficiency at scale.
- **Economic Impact:** Reduces fleet acquisition costs by 60% through payload modularity; eliminates manual labor for solar panel cleaning and emergency logistics coordination.



Future Scope:

How can the idea be developed further?

- Expand payload suite to include thermal imaging for search-and-rescue, pesticide spraying for agriculture, and structural inspection modules for infrastructure monitoring.
- Multi-drone swarm coordination using distributed ML dispatch for simultaneous multi-site operations.
- Autonomous warehouse logistics, last-mile delivery in remote areas, environmental monitoring, and precision agriculture.

Scalability:

Multi-drone swarm coordination using distributed ML dispatch for simultaneous multi-site operations. Licensing platform to drone manufacturers; SaaS dispatch API for emergency services and utility operators.

Future applications:

- Pesticide spraying for agriculture
- Structural inspection for infrastructure monitoring
- Autonomous warehouse logistics
- Last-mile delivery in remote areas
- Environmental monitoring
- Precision agriculture



Meet Team RedFox



Roles & Responsibilities

- Arvind Naraynan M R (Team Lead) – Structural Design, Robotics & Research Head
- M Dharnesh Bala – Impact analysis & Presentation Head
- Govarthan M – Payload Design & Website Head
- Shri Nivas S – Infield Communication & Speech Head

Relevant Skills

Robotics expertise spanning mechanical design, embedded systems, AI/ML, and system integration with proven ideathon competition experience.

			
Arvind Naraynan M R Team Lead Structural Design, Robotics & Research Head	M Dharnesh Bala Presentation Head Impact analysis & presentation strategy	Govarthan M Website Head Payload Design & digital implementation	Shri Nivas S Speech Head Infield communication & public speaking

*Students From Mech D/ B2/ 1st Year

Relevant Skills

Full-stack robotics expertise spanning mechanical design, embedded systems, AI/ML, and system integration with proven ideathon competition experience.